

Final Verification

Independent verification of the implemented system against the handbook §8.3 acceptance checks, the Deliverable 4 specification, and the measured Deliverable 3 baseline.

System Mnemo — Conversational Memory Intelligence System

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Verdict PASS — 8/8 executable checks, 1 reported UNVERIFIED

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Evidence verification/results/ · test_plan.md · security_report.md

Substrate pure-stdlib + sqlite3 (D5-DR-001, D6-DR-002)

VERIFICATION STANCE

Verification was performed **from the specification**, not by accepting the implementation's own assessment. `verification/verify.py` re-derives every §8.3 acceptance check independently and never reads the gate scripts' verdicts. Every claim below is backed by a command whose exit code is reproducible, or it is listed as **UNVERIFIED**. There is no third category. Known failures and residual risks are reported rather than hidden.

1. Headline result

The implemented system passes **11 of 11** adversarial queries on the fixed evaluation set that the naive baseline passed **0 of 11** — and it does so by changing *what is allowed to be a candidate*, not by re-weighting a ranker.

METRIC (SAME FIXED SET, K=5, BUDGET=2000)	NAIVE BASELINE (D3)	MNEMO (D6)
overall_accuracy	0.000 (0/11)	1.000 (11/11)
recall_at_k	0.333	0.778
supersession_failure_rate (F4)	0.800	0.000
inversion_failures (F7)	1	0
cross_tenant_leak_queries (F10)	7 of 11	0
pii_exposure_count (F11)	3	0
coldstart_abstention_failure_rate	1.000	0.000

2. §8.3 acceptance checks — verdicts

#	CHECK	INDEPENDENT EVIDENCE	VERDICT
1	Relevant memories outrank plausible distractors	No distractor precedes its gold on any of the 11 queries. Stronger outcome: distractors are removed upstream (validity / isolation / PII), so they cannot rank at all.	PASS
2	Stale or superseded preferences do not override current	m001/m003/m005/m007 are not current; the current fact is the one injected for q01/q03/q04; a wrongly-invalidated fact is recoverable from history.	PASS
3	No cross-tenant memory under adversarial queries	0 foreign rows across all queries <i>and</i> exact-text adversarial queries for m101/m102; static check confirms no repository method accepts a tenant parameter.	PASS
4	Deletion removes from source storage and retrieval within the window	Delete→requery returns nothing; raw SQL shows 0 rows in <code>memory</code> <i>and</i> <code>memory_embedding</code> ; <code>window_ms</code> $\approx$ 0.11 ms vs the ADR-004 24 h backstop.	PASS
5	Context selection respects the token budget	Budgets 40 / 120 / 2000 all honoured; measured injected tokens $\leq$ budget in each case.	PASS
6	Sensitive-data policy: positive <i>and</i> negative cases	4/4 positive blocked with no raw span retained; 4/4 negative admitted; across all 44 memories the gate blocks exactly the one flagged memory.	PASS
7	Benchmark compared with the naive baseline	Table in §1 plus the 3-arm comparison below.	PASS
+	Memory-borne prompt injection (T4/R4, project-specific)	8/8 overt injections blocked, benign control admitted, 3 subtle survivors quantified .	PASS with documented residual

3. The 3-arm comparison — what actually fixes the failures

Arm 2 exists to falsify the cheap fix (“just prefer newer memories”). It does not survive contact with the data.

METRIC	1 · NAIVE	2 · +RECENCY	3 · VALIDITY-FILTER
overall_accuracy	0.0	0.0	1.0
supersession_failure_rate	0.8	0.8 (unchanged)	0.0
cross_tenant_leak_queries	7	10 (worse)	0
pii_exposure_count	3	3	0
coldstart_abstention_failure_rate	1.0	1.0	0.0

TWO FINDINGS I EXPECTED TO GO THE OTHER WAY

Recency did not move supersession at all (0.80 → 0.80). The *superseded* memory is the one that lexically matches the query — “CRM platform” appears in the stale Salesforce chunk, while the current fact says “migrated to HubSpot” — so a recency bonus competes with relevance instead of overriding it. **Recency made cross-tenant leakage worse** (7 → 10 of 11 queries), because the foreign-tenant near-duplicates are *newer* than the tenant's own facts. This reproduces the Deliverable 3 prediction — a stronger ranking signal makes F10 worse, not better — as a measurement rather than an assertion. Together these settle first-principles falsifiers 1–2: supersession is not a recency-ranking problem, and the isolation / PII / cold-start failures are structural.

4. A defect this verification found in earlier work

Reporting this because it changes how an earlier claim should be read. While building the consolidation path (M4), gate G3 passed but reported an implausible number: a 12-turn conversation thread had only 1 source to consolidate. Investigation showed the write-path supersession rule was firing on every (account, subject) collision, and conversation turns had defaulted to the fact type — so they superseded one another. **31 of 40 admitted memories were being silently invalidated.** That is F5 (loss) committed by the write path.

The consequence for Deliverable 5: its read-path gate had been passing against a corpus of only 9 surviving facts, i.e. the filler noise the ranker was supposed to prove it could handle had already been deleted by the defect. **The gate was passing partly for the wrong reason.** After scoping supersession to slot-valued types only (fact , preference ; events accumulate), the corpus is 30 current facts and **the read-path gate still passes 11/11** — a materially stronger result than before, now for the right reason. recall_at_k fell from 1.0 to 0.778, which is the honest cost of restoring the noise.

5. Residual risks — carried, not closed

ID	RISK	STATUS	MEASURE	NEXT STEP
R4	Subtle indirect-authority injections survive admission; one reaches an injected context	OPEN	3/3 subtle survive; rt09 verified reachable	Provenance-weighted trust; render memories as quoted third-party data
R-P1	Postgres RLS defence-in-depth not executed	UNVERIFIED	—	Run postgres_schema.sql on a real Postgres; exact command in test_plan.md §4
R-P2	Spelled-out identifiers evade the deterministic PII floor	OPEN	evasion reproduced	Presidio ML layer behind the floor (ADR-003)
R-P3	Subject-abbreviation queries (dm) abstain instead of answering	OPEN (fails safe)	recall_at_k 0.778	Subject/query normalization
R6	Op-selector cost unproven	DEFERRED	not built in v1	Remains a prototype per D4

6. Reproducing this verification

```
cd Deliverable-1
bash verification/run_all.sh           # captures every artifact under verification/results/
```

Python 3.8+ standard library only — no install, no network, no services. Fixed dataset (SEED=20260722), fixed k=5 and budget=2000, no randomness at evaluation time; the 3-arm failure metrics are asserted byte-identical across two runs. Only wall-clock figures (window_ms , latency) vary between runs.

VERDICT

PASS. All 8 executable acceptance checks pass across 38 evaluation cases, with 36 unit and integration tests green and gates G0–G4 all exiting 0. One item (Postgres RLS) is reported **UNVERIFIED** rather than assumed, and four residual risks are carried with measured numbers and named next steps. The repository can be set up and tested from its documentation, and every critical claim is backed by reproducible evidence.